

Gioia Zheng

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github.com/GioiaZheng

Profile

Bachelor's student in Applied Computer Science and Artificial Intelligence at Sapienza University of Rome, focused on **Information Retrieval**, **Retrieval-Augmented Generation**, and **scalable backend systems**. Comfortable across the stack — from sparse/dense retrieval pipelines and reranking to API design and reproducible evaluation infrastructure.

Education

Sapienza University of Rome

Sep 2022 – Present

B.Sc. in Applied Computer Science and Artificial Intelligence

Experience

NLP Research Intern

Feb 2026 – Present

Microsoft (Remote)

- Built an open-domain retrieval-augmented QA pipeline on MS MARCO combining sparse (BM25) and dense (FAISS) retrieval with cross-encoder reranking
- Designed an evaluation framework with **paired-bootstrap confidence intervals over 6,980 dev queries**, quantifying reranker lift over the BM25 baseline at the question level
- Introduced **BERTScore as a semantic-proxy metric** on top of exact-match for generation quality, and a regression-failure taxonomy for systematic error analysis
- Built resume-safe runner infrastructure and CI for reproducible long-running indexing and evaluation jobs

Web Development Intern

Mar 2026 – Present

Idearia, Rome, Italy

- Developed and maintained PHP/WordPress backend systems for client-facing websites
- Designed n8n automation workflows to integrate WordPress content and forms with external services

Programming Intern

May 2021 – Jun 2021

Sandbox S.r.l., Italy

- Contributed to a 3D simulation platform and backend tooling

Projects

Retrieval-Augmented QA System

github.com/GioiaZheng/msmarco-genqa

- Open-source RAG retrieval pipeline; modular architecture supporting plug-in retrievers and rerankers
- Reproducible CI, resume-safe runners, and packaging for long-running indexing/evaluation experiments
- Stack: Python, PyTorch, FAISS, Hugging Face Transformers, scikit-learn

Full-Stack Chat System

github.com/GioiaZheng/go-chat-system

- Full-stack chat application; Go backend with concurrent request handling and SQLite persistence
- REST APIs documented in OpenAPI; Vue.js frontend embedded into the Go binary at build time for single-artifact deployment

CiboCompass Mobile App

github.com/GioiaZheng/CiboCompass

- Cross-platform React Native app helping international students navigate Italian restaurant menus
- Offline-first rating system with local cache and backend sync; Go REST API with SQLite persistence

Handwritten OCR System

github.com/GioiaZheng/handwritten-ocr-system

- End-to-end handwritten OCR pipeline in PyTorch on the IAM handwriting dataset
- CNN feature extractor with bidirectional LSTM and CTC alignment for variable-length transcription; evaluated with CER and WER on held-out splits

Skills

Programming

Python, Go, JavaScript, PHP, Java, SQL

ML & NLP

PyTorch, Hugging Face Transformers, RAG, Deep Learning

Information Retrieval

BM25, FAISS, sparse + dense retrieval, cross-encoder reranking, BERTScore, paired-bootstrap evaluation

Backend & Systems Tools

REST APIs, OpenAPI, SQLite, system design, WordPress

Languages

English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)

Additional

WISWUP — 1st Place

Sep 2021

Mensa Italia Member

Dec 2025 – Present